Commodities vs Equities as an Inflation Hedge

Data and Methodology

The analysis covers the period from February 2006 to September 2026. Commodities are represented by the S&P GSCI Commodity Total Return Index (GSCI), and equities by the SPDR S&P 500 ETF (SPY) using adjusted prices.

Monthly real returns were approximated as: Real Return ≈ Nominal Return − Inflation

We classified months in the top 20% of year-over-year CPI inflation as high inflation (>3.64%), while the remaining months were classified as normal inflation. This resulted in 49 high-inflation months and 197 normal-inflation months.

For each asset and regime, we calculated mean monthly real return, volatility, and inflation beta. The beta was estimated using OLS: Asset Return = $\alpha + \beta \cdot$ Inflation + ε

Results

Table 1. Performance by Inflation Regime

Regime	Asset	Mean Real Return (%)	Volatility (%)	Inflation Beta	Beta SE
High Inf.	**GSCI**	−0.35	7.52	10.43	2.74
High Inf.	**SPY**	−1.13	5.45	3.89	2.18
Normal Inf.	**GSCI**	+0.04	6.30	11.38	1.48
Normal Inf.	**SPY**	+1.23	3.91	1.66	1.03

Figure 1. Mean Monthly Real Returns by Inflation Regime

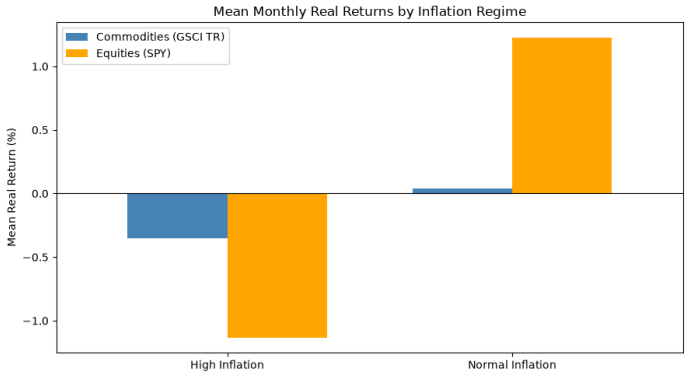
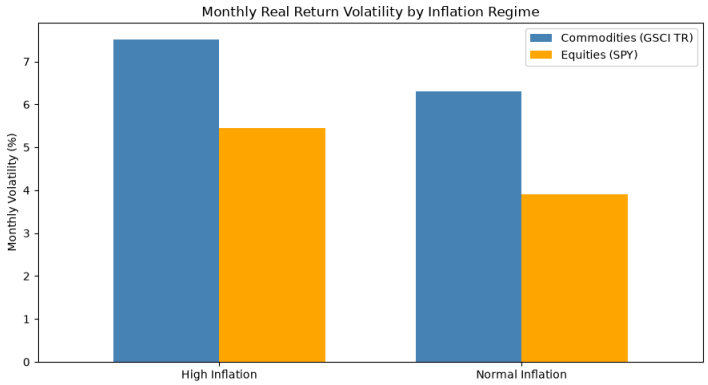


Figure 2. Monthly Real Return Volatility by Inflation Regime



During high inflation, GSCI outperformed SPY by 0.78 points per month (-0.35%- (-1.13%)). However, both assets had negative average real returns, meaning neither preserved purchasing power but provided relative hedging.

During normal inflation, SPY performed better, earning 1.23% per month compared with GSCI's 0.04%.

GSCI showed higher volatility in both regimes (high-inflation: GSCI 7.52 vs. SPY 5.45 / normal-inflation: GSCI 6.30 vs. SPY 3.91).

GSCI's inflation betas were positive and statistically significant at the 5% level in both regimes (10.43 and 11.38). SPY's betas were not statistically significant (3.89 and 1.66). This shows stronger inflation sensitivity for commodities.

Discussion

Of the 49 high-inflation months, 28 occurred during 2021–2023 (after the pandemic). To test the sensitivity of our findings, we excluded these years while keeping the inflation threshold (>3.64%).

Table 2. Robustness Check — High-Inflation Mean Real Returns

Sample	GSCI (%)	SPY (%)
Full sample (49 months)	−0.35	−1.13
Excluding 2021–2023 (21 months)	−2.11	−2.22

When we exclude, GSCI's advantage declined from 0.78 to 0.11 (-2.11-(-2.22)). This shows that the results heavily depend on the historical periods.

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